Temporal Metamorphic Testing for RAG-Based LLMs under Evolving Knowledge

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***Abstract*—Evaluating the temporal consistency of Retrieval-Augmented Generation (RAG) systems is challenging because conventional evaluation methods rely on ground-truth answers, which are difficult to define under continuously evolving knowledge. This paper presents a temporal metamorphic testing framework for RAG-based large language models (LLMs) that evaluates relational consistency across multiple executions over time rather than comparing outputs against static reference answers. To support temporal metamorphic testing, we construct controlled temporal revision families from seed facts extracted from real-world documents. Each fact is systematically cloned and transformed across multiple timestamps to simulate evolving knowledge states. Based on this design, temporal metamorphic relations are instantiated to systematically test RAG systems under different knowledge snapshots. Experiments on 2344 metamorphic test instances show that 48.04% of them violate the expected temporal relations, indicating that temporal inconsistencies are widespread in the evaluated RAG system. These results demonstrate that temporal metamorphic testing provides an effective way to assess temporal reliability in RAG systems without requiring explicit ground-truth answers.**

***Index Terms*—Retrieval-Augmented Generation(RAG), Temporal Reasoning, Metamorphic Testing(MT), Temporal Consistency**

I. Introduction

Large Language Models (LLMs) have demonstrated strong capabilities in a wide range of natural language processing (NLP) tasks [1], [2]. However, their knowledge is inherently static, reflecting the state of the training data at the time the model was trained. This limitation becomes particularly problematic for time-sensitive queries, where factual correctness depends on temporal context [3], [4], [5]. Retrieval-Augmented Generation (RAG) has been proposed to address the issue by incorporating external knowledge sources during inference [6], [7]. By retrieving relevant documents from an updatable corpus, RAG systems can provide more up-to-date and grounded responses. Despite this advantage, recent studies have shown that RAG systems still struggle with temporal consistency, often producing outdated, inconsistent, or temporally misaligned answers [8], [9].

Evaluating the temporal behavior is inherently challenging. Traditional evaluation methods rely on predefined ground-truth answers, which are difficult to obtain for evolving knowledge. Metamorphic Testing (MT) [10], offers a promising alternative by examining relations among multiple executions rather than relying on an explicit oracle [11], [12], [13]. While recent work has applied MT to detect factual hallucinations in LLMs, limited attention has been given to temporal inconsistencies in RAG systems [8].

To address this challenge, this paper proposes a temporal metamorphic testing framework for RAG systems. Instead of relying on static ground-truth answers, the framework evaluates whether outputs generated under different temporal conditions satisfy expected relational constraints. To support such testing, we construct a temporally aware evaluation setting based on time-stamped knowledge snapshots and instantiate a set of temporal metamorphic relations (MR) [14], [15] for executable testing. Experiments on 2344 test instances show that 48.04% of them violate the expected temporal relations, suggesting that temporal inconsistencies are widespread in RAG systems.

The main contributions of this work are as follows:

- We propose an *oracle-free temporal metamorphic testing framework* for RAG systems that detects temporal hallucinations through temporal relation violations rather than factual answer correctness.
- We introduce *Temporal Revision Families (TRFs)*, where seed facts extracted from real-world documents are cloned and mutated across timestamps to form controlled evolving knowledge states for temporal testing.
- We define four Temporal Metamorphic Relations (TMRs) that evaluate snapshot stability, update sensitivity, historical grounding, and event ordering preservation in temporally evolving knowledge environments.
- We implement an end-to-end testing pipeline and empirically investigate temporal inconsistency patterns in RAG systems under controlled temporal evolution.

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II. Background

A. Retrieval-Augmented Generation

RAG extends LLMs by incorporating external knowledge sources at the stage when the model generates answers to input queries [6], [7]. In a typical RAG pipeline, relevant documents are retrieved from a knowledge base and provided as context for response generation [6]. The model then produces a final answer conditioned on both the input query and the retrieved evidence.

This approach improves factual grounding and allows the system to access more up-to-date information. However, despite these advantages, RAG systems still face challenges in maintaining temporal consistency, particularly when the retrieved documents correspond to a different temporal context than that of the query.

B. Temporal Question Answering

Temporal question answering focuses on queries whose answers depend on time, such as "Who was the president of the United States in 2010?". Recent work has introduced datasets for evaluating temporal reasoning in QA and RAG systems, including ChronoQA [8] and other time-sensitive QA benchmarks [16], [2].

These datasets highlight the importance of temporal alignment between queries and retrieved knowledge.

C. Hallucination in LLMs

Hallucination refers to the phenomenon where language models generate incorrect or unsupported information [17]. In RAG systems, hallucination may occur due to retrieval errors, outdated documents, or inconsistencies between retrieved evidence and generated responses [18], [9]. Temporal hallucination is a specific form of the problem, where the generated answer is temporally incorrect, even if it may have been valid at another point in time.

D. Metamorphic Testing

Metamorphic Testing (MT) [10] is a software testing methodology designed for systems where expected outputs are difficult to determine [11], [12], [13]. Instead of relying on ground-truth outputs, MT checks whether expected relationships hold across multiple executions under systematically transformed inputs.

Recent work has applied MT to LLMs for detecting hallucination and inconsistency issues [19], [20]. However, existing approaches primarily focus on factual inconsistencies and do not explicitly address temporal behavior in RAG systems.

These limitations highlight the need for systematic methods to evaluate temporal behavior in RAG systems, which we address in this work.

III. Methodology

Unlike traditional evaluation methods that rely on static ground-truth answers, our approach models temporal consistency as relational constraints across multiple executions, making it more suitable for evaluating systems under evolving knowledge conditions.

A. System Formulation

Let $\mathcal{Q}$ denote the query space, $\mathcal{S}$ the set of temporally indexed knowledge snapshots, where each knowledge snapshot represents a time-stamped version of the corpus containing only the information available up to a specific time point, and $\mathcal{Y}$ the output space of the RAG system. The system under test is modelled as a temporal RAG function

$$F : \mathcal{Q} \times \mathcal{S} \to \mathcal{Y}, \tag{1}$$

where

$$F(q, s_t) \tag{2}$$

denotes the final answer generated for query q when the RAG pipeline is executed over the knowledge snapshot s_t at time t.

Let K_t denote a temporal knowledge snapshot and let $\mathcal{R}_i$ denote the i-th TRF. Each TRF consists of a sequence of timestamped revision tuples:

$$\mathcal{R}_i = \left\{ \tau_i^{t_1}, \tau_i^{t_2}, \ldots, \tau_i^{t_n} \right\}. \tag{3}$$

The expected temporal relations are derived from the internal structure of $\mathcal{R}_i$ rather than from real-world ground-truth answers.

B. Temporal Input Transformations

Two forms of temporal transformation are considered.

First, snapshot transformation changes the temporal state of the knowledge base while keeping the query unchanged:

$$T_s : (q, s_t) \mapsto (q, s_{t'}). \tag{4}$$

Second, query transformation modifies the explicit or implicit temporal condition in the query while keeping the same underlying system interface, meaning that the same model and retrieval pipeline are used across all transformed queries.

$$T_q : (q, s_t) \mapsto (q', s_t), \tag{5}$$

where q' is a temporally transformed variant of q, such as replacing "current" with "in 2019" or anchoring the question to a fixed historical time.

C. Temporal Metamorphic Relations

Our design is inspired by prior work on MT [10], [19]. Each Temporal Metamorphic Relation (TMR) consists of a temporal input transformation and an expected relation among outputs.

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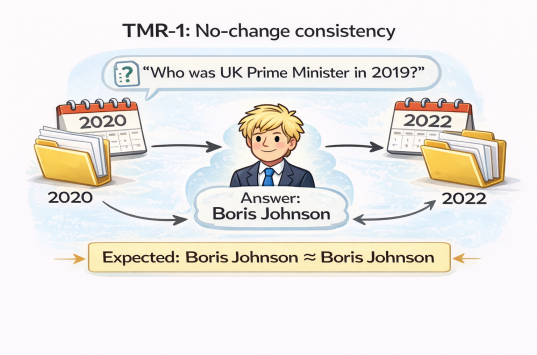


Fig. 1. Illustration of TMR-1 (No-Change Consistency). The answer to the query "Who was UK Prime Minister in 2019?" remains Boris Johnson across different temporal snapshots (e.g., 2020, 2022, 2025), since the underlying fact does not change. The figure is generated using an AI-based image generation tool and adapted by the authors.

1) TMR-1: Snapshot Stability: For a stable TRF, the underlying value remains unchanged across temporal snapshots:

$$v_i^{t_1} = v_i^{t_2} = \ldots = v_i^{t_n}, \tag{6}$$

where $v_i^{t_j}$ denotes the value associated with the i-th revision family at timestamp t_j.

For a non-temporal query q, the generated outputs are expected to remain semantically equivalent across all snapshots:

$$F(q, K_{t_1}) \approx F(q, K_{t_2}) \approx \ldots \approx F(q, K_{t_n}). \tag{7}$$

A violation of this relation indicates temporal instability, where the system produces inconsistent outputs despite no change in the underlying knowledge state.

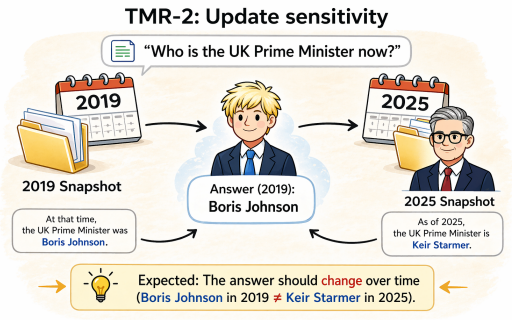


Fig. 2. Illustration of TMR-2 (Update Sensitivity). The answer to the query "Who is the UK Prime Minister now?" changes across snapshots, reflecting real-world updates (e.g., Boris Johnson in 2019 versus Keir Starmer in 2025).

2) TMR-2: Update Sensitivity: For an evolving TRF, the associated value changes across temporal snapshots:

$$v_i^{t_a} \neq v_i^{t_b}, \tag{8}$$

where t_a and t_b correspond to different knowledge snapshots.

For a present-time query q_{now}, the generated outputs should reflect the updated knowledge state:

$$F(q_{\text{now}}, K_{t_a}) \not\approx F(q_{\text{now}}, K_{t_b}). \tag{9}$$

A violation of this relation indicates stale retrieval or temporal insensitivity, where the system fails to update its response according to the evolving knowledge state.

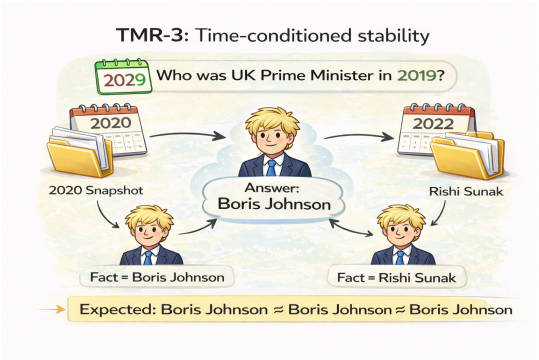


Fig. 3. Illustration of TMR-3 (Time-Conditioned Stability). For a time-anchored query such as "Who was UK Prime Minister in 2019?", the system should return the historically correct answer (Boris Johnson), even when later snapshots contain updated information, thereby avoiding temporal leakage.

3) TMR-3: Historical Grounding: For a query explicitly anchored to a historical timestamp τ,

$$q_\tau, \tag{10}$$

the expected answer is determined by the value stored in the revision family at timestamp τ:

$$A(q_\tau) = v_i^\tau. \tag{11}$$

Therefore, regardless of which future snapshot is used during retrieval, the generated outputs should remain consistent:

$$F(q_\tau, K_{t_1}) \approx F(q_\tau, K_{t_2}) \approx \ldots \approx F(q_\tau, K_{t_n}) \approx v_i^\tau, \tag{12}$$

where

$$\tau \leq t_1 \leq t_2 \leq \cdots \leq t_n. \tag{13}$$

A violation of this relation indicates temporal leakage, where information introduced in later revisions incorrectly influences answers to historically anchored queries.

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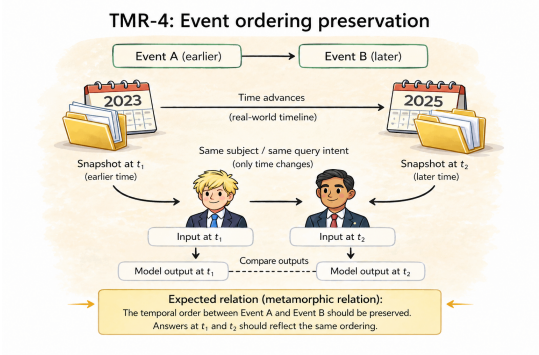


Fig. 4. Illustration of TMR-4 (Event Ordering Preservation). The temporal order between events (e.g., Boris Johnson preceding Rishi Sunak) should remain consistent across different snapshots.

4) TMR-4: Event Ordering Preservation: For two events e_a and e_b within a TRF, if their temporal order remains unchanged across all relevant snapshots,

$$time(e_a) < time(e_b), \quad \forall t \in \mathcal{T}, \tag{14}$$

then the generated outputs should preserve the same ordering relation:

$$\text{order}\big(F(q, K_{t_1})\big) = \text{order}\big(F(q, K_{t_2})\big) = \cdots = \texttt{before}. \tag{15}$$

A violation of this relation indicates incorrect temporal reasoning, where the system fails to preserve the temporal order encoded in the underlying knowledge state.

D. Drift Quantification

Let N be the total number of MT instances, where each MT instance corresponds to a concrete test case derived from a temporal metamorphic relation, consisting of a source query and its associated follow-up queries across different knowledge snapshots, and let $v_i \in \{0, 1\}$ indicate whether the ith instance violates its expected TMR or not. A basic drift score is defined as

$$D = \frac{1}{N} \sum_{i=1}^{N} v_i. \tag{16}$$

This score measures the proportion of temporal metamorphic violations over all executed tests [21]. A larger D indicates lower temporal reliability of the RAG system.

More generally, to account for varying degrees of violation severity, the score can be extended to a weighted form:

$$D_w = \frac{\sum_{i=1}^{N} w_i \delta_i}{\sum_{i=1}^{N} w_i}, \tag{17}$$

where δ_i represents the magnitude of violation based on semantic divergence or inconsistency), and w_i is a task-specific weight. This formulation allows the incorporation of graded evaluation signals, such as semantic similarity, contradiction strength, or temporal misalignment.

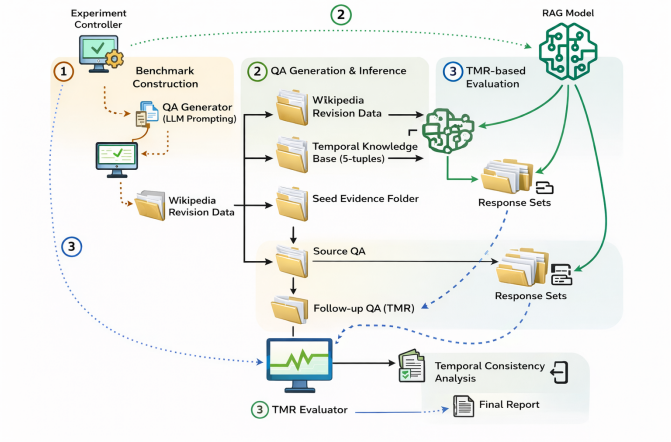


Fig. 5. Overview of the proposed framework

IV. Framework Implementation

Figure 5 illustrates the implementation of the proposed temporal metamorphic testing framework.

The pipeline consists of five stages: seed document collection, seed tuple extraction, temporal revision family construction, TMR-based test generation, and RAG-based metamorphic evaluation.

A. Seed Document Collection

We use real-world documents as seed sources rather than as ground-truth references. In our implementation, news articles are collected from public web sources. Each article provides natural language context, entities, temporal expressions, and event descriptions that can be used to construct seed facts.

The role of these documents is not to define factual correctness. Instead, they provide diverse and realistic linguistic material for generating temporal testing units. This design preserves the naturalness of retrieval contexts while avoiding dependence on complete historical revision records.

B. Temporal Revision Family Construction

For each extracted seed tuple, we construct a TRF as defined in Section III-A. A TRF is generated by cloning the seed tuple across multiple timestamps and applying controlled temporal mutations to simulate evolving knowledge states.

The value associated with each revision may remain unchanged or be systematically modified according to the assigned temporal pattern. We consider four temporal patterns: stable, evolving, historical, and event-ordering, corresponding to the four TMRs.

For stable TRFs, the value remains unchanged across snapshots. For evolving TRFs, the value changes over time to simulate knowledge updates. For historical TRFs, later snapshots may contain conflicting future values, allowing the framework to test whether the system correctly respects historical temporal anchors. For event-ordering TRFs, pairs of events are generated with predefined temporal relationships to evaluate temporal reasoning consistency.

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C. Snapshot Corpus Construction

After TRFs are generated, we construct temporally indexed snapshot corpora. For each timestamp t_j, a knowledge snapshot K_{t_j} is built by collecting the corresponding revision evidence from all TRFs.

Each revision tuple is converted into a natural language evidence sentence, such as:

As of 2021, the CEO of Company X was Person Y.

These evidence sentences are then used as retrievable documents in the RAG corpus. This step allows the RAG system to operate over natural language evidence rather than symbolic tuples, while the underlying temporal relations remain controlled by the TRF structure.

D. TMR-based Test Case Generation

For each TRF, we automatically generate metamorphic test cases according to its temporal pattern.

For TMR-1, stable TRFs generate non-temporal queries whose outputs should remain equivalent across snapshots. For TMR-2, evolving TRFs generate non-temporal queries whose outputs should change when the underlying value changes. For TMR-3, historical TRFs generate time-anchored queries whose outputs should remain grounded to the value at the specified historical timestamp. For TMR-4, event-ordering TRFs generate ordering queries that require the system to preserve the predefined temporal order between events.

Each test case consists of a source execution and one or more follow-up executions:

$$\{F(q_0, K_{t_0}), F(q_1, K_{t_1}), \ldots, F(q_m, K_{t_m})\}. \tag{18}$$

The test oracle is the expected relation among these outputs, not a real-world answer.

E. RAG Configuration

Given a query q and a temporal snapshot K_t, the retriever identifies the top-k most relevant evidence chunks from the snapshot corpus. The retrieved chunks are then provided to the generator to produce the final answer:

$$y = F(q, K_t). \tag{19}$$

The same retrieval and generation pipeline is used across all source and follow-up executions. This ensures that observed violations are caused by temporal inconsistency rather than changes in system configuration.

F. Metamorphic Evaluation

The generated outputs are evaluated by checking whether they satisfy the expected TMR. For equivalence-based relations, we use semantic similarity or LLM-based judgment to determine whether two outputs express the same value. For update-sensitive relations, we check whether outputs change consistently with the TRF-defined value changes. For historical grounding, we verify whether outputs remain aligned with the value at the anchored timestamp. For event ordering, we verify whether the predicted ordering matches the predefined order in the TRF.

The overall violation rate is computed as:

$$D = \frac{1}{N} \sum_{i=1}^{N} v_i, \tag{20}$$

where N is the number of metamorphic test instances and $v_i \in \{0, 1\}$ indicates whether the i-th instance violates its expected temporal metamorphic relation.

V. Experimental Results

We evaluated the proposed framework on a temporally constructed dataset consisting of 6259 temporal revision instances. From these instances, 2344 metamorphic test cases were generated according to the four defined TMRs. Each test case contains a source execution and one or more follow-up executions derived from the same TRF.

The experiments were conducted using a RAG system instantiated with the Qwen2.5-3B model [22]. For each test case, the generated outputs were compared against the expected temporal relation defined by the corresponding TRF.

A. Evaluation Method

For each metamorphic test instance, we compare the generated outputs against the expected relation specified by the corresponding TMR.

To quantify semantic consistency, both the expected answer and the generated answer are encoded using the `all-MiniLM-L6-v2` sentence embedding model. Let $E(y)$ denote the embedding representation of answer y.

The similarity between two answers is computed using cosine similarity:

$$Sim(y_1, y_2) = \frac{E(y_1) \cdot E(y_2)}{\|E(y_1)\| \|E(y_2)\|}. \tag{21}$$

A similarity threshold $\theta = 0.85$ is used to determine whether the expected temporal relation is satisfied.

B. Overall Performance

The overall evaluation results are summarized in Table I.

TABLE I
Summary of Temporal Metamorphic Testing Results

Metric	Value
Temporal Revision Instances	6259
TMR Test Cases	2344
Satisfaction Rate	51.96%
Violation Rate (D)	48.04%

The results indicate that 48.04% of the constructed metamorphic test cases violated their expected temporal relations. This value corresponds to the temporal drift score D defined in Section III and suggests that temporal inconsistencies are common in the evaluated RAG system under evolving knowledge conditions.

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Although the retrieval corpus contains temporally structured information, the model frequently fails to preserve the temporal relations encoded in the Temporal Revision Families. These violations demonstrate that access to temporal knowledge alone does not guarantee temporally consistent behavior.

C. TMR-wise Observations

Test instances were constructed for all defined temporal metamorphic relations, including TMR-4 (Event Ordering Preservation). In TMR-4 cases, the system was required to preserve the correct temporal order between two events across different snapshots. Violations of this relation were also observed, indicating that temporal inconsistency in RAG systems can affect not only single-fact predictions but also relational reasoning over event sequences.

D. Analysis of Temporal Violations

We further analyzed the types of temporal inconsistencies observed in the evaluation.

Stale Answer Generation. In many cases, the system failed to update its answer when the underlying fact changed across temporal snapshots. This behavior reflects violations of update-sensitive relations and suggests that the retrieval component may return outdated information.

Temporal Instability. For queries whose answers should remain unchanged over time, the system produced inconsistent responses across different snapshots. This indicates instability in retrieval or generation, even when the TRF-defined temporal state is invariant.

Temporal Leakage. The system occasionally drew from information available at later times for time-conditioned queries such as queries that reference previous time points. This indicates that the model doesn't completely adhere to the temporal constraints indicated in the query.

E. Implications

The observed violations suggest that temporal failures in RAG systems arise from both retrieval and generation. Some errors stem from outdated or misaligned retrieved evidence, while others reflect the model's inability to reason consistently under temporal constraints. These findings indicate that temporal reliability in RAG systems depends not only on access to updated knowledge, but also on whether the system can correctly align retrieved evidence with the time condition expressed in the query.

VI. Conclusion

This paper presents a temporal metamorphic testing framework for evaluating RAG systems under evolving knowledge conditions. The results show that temporal inconsistencies are widespread, with 48.04% of the tested instances violating expected temporal relations. These findings indicate that static answer-based evaluation is insufficient for assessing temporal reliability in RAG systems. Instead, relational validation across temporally transformed executions provides a more suitable way to expose temporal weaknesses in retrieval and generation.

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